

Wearable Robotics

L12 - The Role of Series and Parallel Elasticity

Prof. Bolívar

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Announcement:

Midterm exam after break
(take-home)



No More Slow Flexible Robots:
Discussions on advancements in
design and control of compliant
robotic systems

- Clutching using magnetorheological fluid
- Series compliance as part of the leg design

The role of compliance in series and parallel with the motor

Scenario:

- We know the load requirements

(Task)

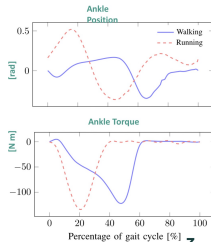
Kinematics

Inverse kinematics

Kinetics

Inverse dynamics

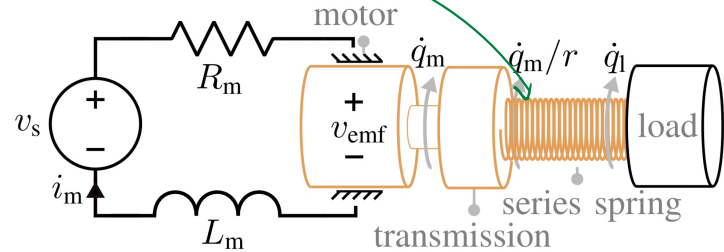
For example,



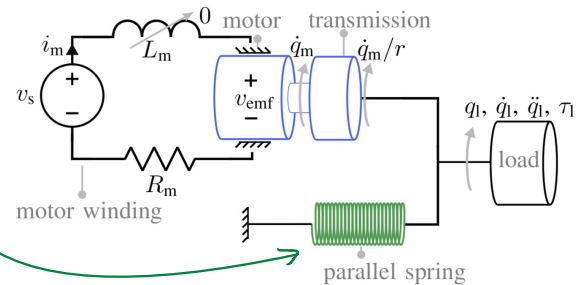
- We want to select the electric motor and transmission to accomplish the task

- How would a parallel or series spring affect the electric motor and transmission selection?

Series Elastic Actuator (SEA)

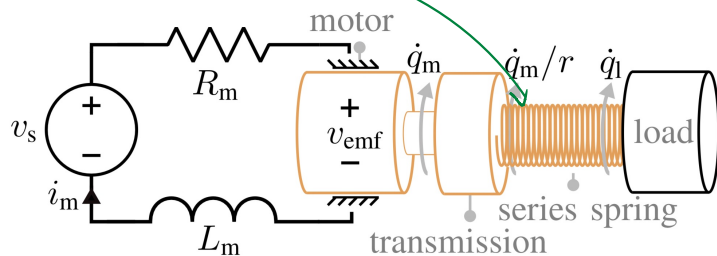


Parallel Elastic Actuator (PEA)

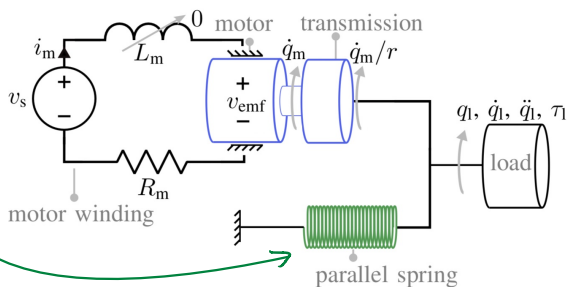


Intuition

Series Elastic Actuator (SEA)



Parallel Elastic Actuator (PEA)



From previous lectures, the equation that describes the motor dynamics:

$$J_m \ddot{q}_m = \tau_m - b_m \dot{q}_m + \frac{\tau_l}{r}$$

Mass moment of inertia
Electromagnetic torque = $k_t i_m$
damping

q_m : motor position
 r : reduction ratio
 τ_l : load torque

Load and motor kinematics are decoupled

$$i.e., \frac{q_m}{r} \neq q_l$$

Reduce impact load
Modify motor speed potential to reduce motor power

Potential to reduce inertial and damping torques

The spring can work as a soft loadcell

Control load torque by controlling spring deflection
HUGE benefit when working with highly geared systems

Load and motor kinematics are coupled

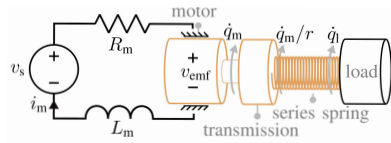
$$\frac{q_m}{r} = q_l$$

Parallel spring and motor have the potential to work together to reduce motor torque

HUGE benefit when working with direct or quasi-direct drives

Effects of series spring in energy consumption

SEA:



Energy consumption of SEA:

(Energy provided by the electrical power supply)

$$\begin{aligned}
 E_m &= \int_{t_0}^{t_f} i_m v_s dt \\
 &= \int_{t_0}^{t_f} i_m (i_m R_m + v_{emf} + L_m \frac{di_m}{dt}) dt \\
 &= \int_{t_0}^{t_f} \left(i_m^2 R_m + \underbrace{\tau_m \dot{q}_m}_{i_m v_{emf}} + L_m i_m \frac{di_m}{dt} \right) dt \\
 &\quad \begin{matrix} i_m = \tau_m k_r^{-1} \\ v_{emf} = \dot{q}_m k_r \end{matrix}
 \end{aligned}$$

$$= \int_{t_0}^{t_f} \left(\frac{\tau_m^2}{k_m^2} + \tau_m \dot{q}_m + L_m i_m \frac{di_m}{dt} \right) dt$$

$k_m = k_r r^2$

Let's analyze periodic motion to gain insight. For example, walking and stair ambulation can be analyzed as periodic motion.

$$\begin{aligned}
 E_m &= \int_{t_0}^{t_f} \left(\frac{\tau_m^2}{k_m^2} + I_m \dot{q}_m \dot{q}_m + b_m \dot{q}_m^2 - \frac{\tau_e \dot{q}_m}{r} \right) dt + \int_{q_0}^{q_f} L i_m dq \\
 &\quad \begin{matrix} 0 \text{ because } i_m = i_f \\ \delta: \text{spring elongation} \\ \tau_e = k_r (q_e - \frac{q_m}{r}) \end{matrix} \\
 &= \int_{t_0}^{t_f} \left(\frac{\tau_m^2}{k_m^2} + b_m \dot{q}_m^2 \right) dt + \int_{t_0}^{t_f} I_m \dot{q}_m \frac{d\dot{q}_m}{dt} dt - \int_{t_0}^{t_f} \tau_e \left(\frac{\dot{q}_m}{r} - \dot{q}_e + \dot{q}_e \right) dt \\
 &\quad \begin{matrix} \text{Energy stored in the spring} \end{matrix} \\
 &= \int_{t_0}^{t_f} \left(\frac{\tau_m^2}{k_m^2} + b_m \dot{q}_m^2 \right) dt + \int_{\dot{q}_{m0}}^{\dot{q}_{mf}} I_m \dot{q}_m d\dot{q}_m + \int_{\dot{q}_e}^{\dot{q}_e} \tau_e \frac{d\delta}{dt} dt - \int_{t_0}^{t_f} \tau_e \dot{q}_e dt
 \end{aligned}$$

$$E_m = \int_{t_0}^{t_f} \left(\frac{\tau_m^2}{k_m^2} + b_m \dot{q}_m^2 \right) dt - \int_{t_0}^{t_f} \tau_e \dot{q}_e dt$$

Energy from the load. It is given by the task and it is independent of the spring stiffness

If the load requires 20J per period the power supply will need to provide those 20J regardless of the spring stiffness

Q: How can the series spring affect the energy consumption?

(The load energy is still required!)

Springs can only reduce energy losses in periodic motion

$$E_m = \underbrace{\int_{t_0}^{t_f} \left(\frac{\tau_m^2}{k_m^2} + b_m \dot{q}_m^2 \right) dt}_{\text{Energy losses}} - \underbrace{\int_{t_0}^{t_f} \tau_e \dot{q}_e dt}_{\substack{\text{Independent} \\ \text{of spring}}}$$

Note that each term is always positive

For simplicity, let's analyze a linear series spring

$\tau_e = k \left(q_e - \frac{q_m}{F} \right)$ → The load torque relates to motor position through the spring equation
 (Stiffness) (Elongation)

$-q_m = r \left(\frac{\tau_e}{k} - q_e \right)$ (1) → It will be simpler (from the optimization perspective) to study the spring effect through its compliance ($1/k$) instead of its stiffness

$q_m = -\alpha r \tau_e + r q_e$
 Spring compliance.

$\dot{q}_m = -\alpha r \dot{\tau}_e + r \dot{q}_e$ → (2) $\Rightarrow \dot{q}_m = a_2 \alpha + b_2$
 $a_2 = -r \dot{\tau}_e$
 $b_2 = r \dot{q}_e$

$\ddot{q}_m = -\alpha r \ddot{\tau}_e + r \ddot{q}_e$ → (3)

The motor torque as a function of spring compliance becomes (using 1 and 2)

$$\begin{aligned}
 \tau_m &= I_m \ddot{q}_m + b_m \dot{q}_m - \tau_e \\
 &= I_m (-\alpha r \ddot{\tau}_e + r \ddot{q}_e) + b_m (-\alpha r \dot{\tau}_e + r \dot{q}_e) - \tau_e \\
 &= \underbrace{(-I_m r \ddot{\tau}_e - b_m r \dot{\tau}_e)}_{a_1} \alpha + \underbrace{(I_m r \ddot{q}_e + b_m r \dot{q}_e - \tau_e)}_{b_r} \\
 &= a_1 \alpha + b_r
 \end{aligned}$$

$$\tau_m^2 = a_1^2 \alpha^2 + 2a_1 b_r \alpha + b_r^2$$

$$\dot{q}_m^2 = a_2^2 \alpha^2 + 2a_2 b_2 \alpha + b_2^2$$

Thus,

$$E_m = \int_{t_0}^{t_f} \left[\left(\frac{a_1^2}{k_m^2} + a_2^2 b_m \right) \alpha^2 + \left(\frac{2a_1 b_r}{k_m^2} + 2a_2 b_2 b_m \right) \alpha + \left(\frac{b_r^2}{k_m^2} + b_2^2 b_m - \tau_e \dot{q}_e \right) \right] dt$$

Energy consumption is a convex quadratic function of series spring compliance

Thus,

$$E_m = \int_{t_0}^{t_f} \left[\left(\frac{a_1^2}{k_m^2} + a_2^2 b_m \right) \alpha^2 + \left(\frac{2a_1 b_1}{k_m^2} + 2a_1 b_2 b_m \right) \alpha + \left(\frac{b_1^2}{k_m^2} + b_2^2 b_m - 2\dot{q}_0 \right) \right] dt$$

$$= a \alpha^2 + b \alpha + c$$

where

$$a = \int_{t_0}^{t_f} \left(\frac{a_1^2}{k_m^2} + a_2^2 b_m \right) dt$$

$$b = \int_{t_0}^{t_f} \left(\frac{2a_1 b_1}{k_m^2} + 2a_1 b_2 b_m \right) dt$$

$$c = \int_{t_0}^{t_f} \left(\frac{b_1^2}{k_m^2} + b_2^2 b_m - 2\dot{q}_0 \right) dt$$

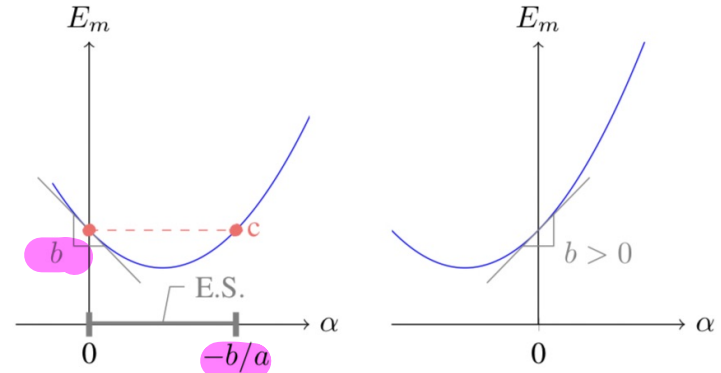


Fig. 2. Left: Energy consumption as a function of compliance, α , where the energy savings (E.S.) region $0 \leq \alpha \leq -b/a$ provides E_m below the rigid level c . Right: Case of motor and load that would not benefit energetically from series elasticity.

* Note: a_1 and a_2 are known constants that depend on the load kinematics and kinetics.

Energy consumption is a convex quadratic function of series spring compliance

$$E_m = a\alpha^2 + b\alpha + c$$

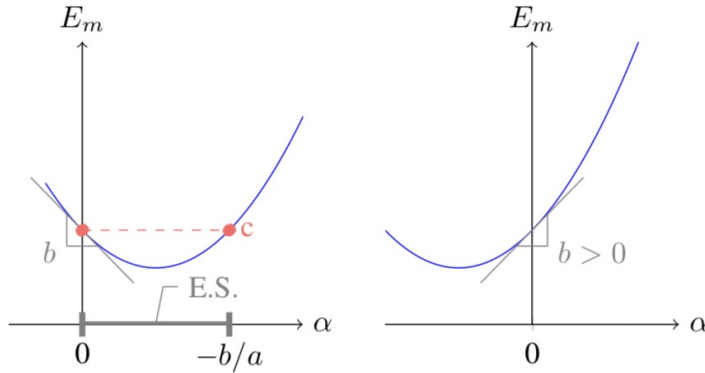


Fig. 2. Left: Energy consumption as a function of compliance, α , where the energy savings (E.S.) region $0 \leq \alpha \leq -b/a$ provides E_m below the rigid level c . Right: Case of motor and load that would not benefit energetically from series elasticity.

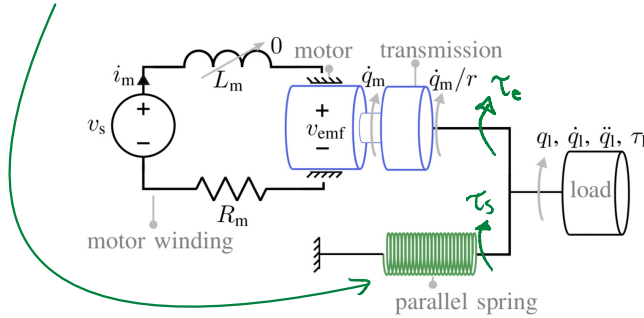
Properties of this convex quadratic function:

- 1) $\frac{d^2 E_m}{d\alpha^2} = 2a \geq 0$; the function is convex
- 2) $\lim_{\alpha \rightarrow 0} E_m = c$; c is the value of energy required to perform the task without spring
- 3) The value of compliance between 0 and $-b/a$ is the energy savings region
- 4) The value of b defines if it is useful to have series compliance

Kahoot!

Effects of parallel springs on RMS motor torque

Parallel Elastic Actuator (PEA)



Dynamic equations:

$$\tau_e + \tau_s + \tau_l = 0 \Rightarrow \tau_e = -\tau_l - \tau_s$$

External torque
on the motor

Torque
Parallel
Spring

Torque load

Parallel spring:

$$\tau_s = -k(q_l + \delta)$$

$$= -kq_l + \tau_{pp}$$

Parallel Spring Preload

k : parallel spring
stiffness

δ : spring preload

$$\tau_m = I_m \ddot{q}_m + b_m \dot{q}_m - \frac{\tau_e}{r}$$

$$= I_m \ddot{q}_m + b_m \dot{q}_m + \frac{\tau_e + \tau_s}{r}$$

Note that $q_m = q_l$. Thus,

$$\tau_m = \underbrace{I_m \ddot{q}_l + b_m \dot{q}_l}_{b} + \underbrace{\frac{\tau_e}{r} - \frac{k}{r} q_l + \frac{\tau_{pp}}{r}}_{a = \begin{bmatrix} -\frac{k}{r} \\ \frac{1}{r} \end{bmatrix}}$$

$$\tau_m = a^T x + b;$$

where

$$x = \begin{bmatrix} q_l \\ \tau_{pp} \end{bmatrix}$$

RMS Torque

$$\text{RMS}(\tau_m) = \sqrt{\frac{1}{t_f - t_0} \int_{t_0}^{t_f} \tau_m^2 dt}$$